Advanced Labs II Report

Vibration of Structures

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Experiment Purpose and Overview

This experiment aims to understand the resonance of structures, conducting vibration tests for some of basic structural mechanics components, a rotating shaft, and plate bending. We identified mode frequencies and mode shapes of each target from experimental data, and understand its relationship to the eigen vale analysis of the structural models. In the first week, we did vibration tests to a rotating shaft, a beam and a plate. We also learned and compared the difference between theoretical model and experimental model. In the second week, we did numerical modal analysis for a plate structure using finite element analysis (FEM).

We did two experiments: vibration of the rotating shaft and vibration of a flat plate.

Experiment 1: Vibration of the Rotating Shaft

a) Experimental task

1. Obtain critical speed from vertical and horizontal vibration waveforms obtained from experiments.
2. Obtain a theoretical solution for the critical speed.
3. Compare experimental results with theotical solutions
4. Consider the difference between vertical and horizontal critical speeds.

The experimental equipment for rotary shaft in Figure 1.

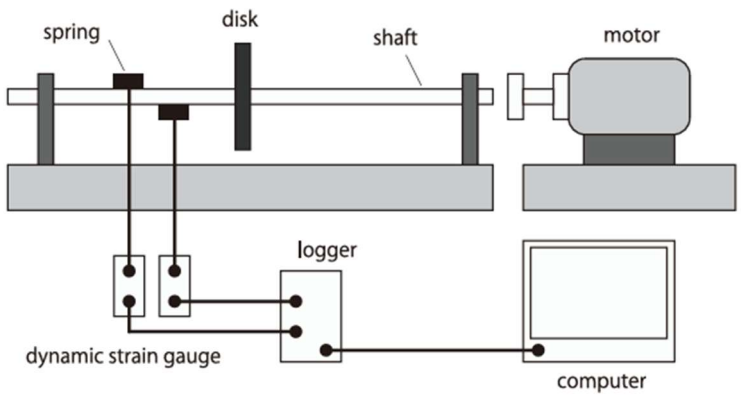


Figure 1 Experimental equipment for rotary shaft

b) Experimental method

1. Couple the rotating shaft to the motor drive shaft and switch on the motor.
2. Accelerate the rotating shaft sufficiently to reach or exceed the critical speed and switch off the motor.
3. The coupling between the motor and the rotary shaft is dislodged and only the rotary shaft rotates.
4. The rotational speed of the rotational shaft gradually decreases and the shaft begins to vibrate when it reaches a critical speed.
5. Vibration is obtained from strain gauges affixed to the holding spring and observed by computer.

The critical speed of each mode is recorded. In the textbook, the critical speed is the period of resonance.

c) The acquired data:

• Sampling period Δt = 50 μs

• Voltage (equivalent to strain)

• Distance from the end to the weight a = 364 mm (from the motor-side end)

From the experiment, we obtained the time history of the rotary shaft and find resonance from rotary shaft. We did two trials. For each trial, two outputs are recorded from the experiment by the computer into CSV file. The horizontal output is V_o1 (horizontal strain) and the vertical output (vertical strain) is V_o2, each plotted against time (μs).

1. First trial

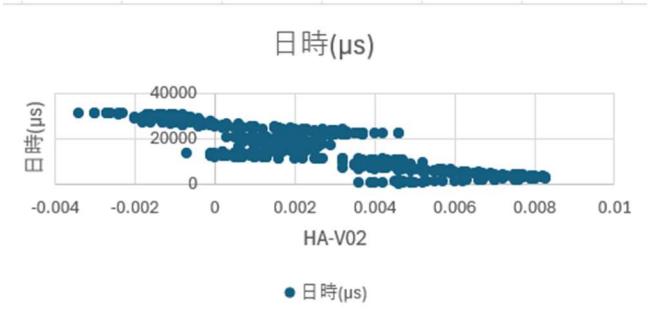


Figure 2 HA-V02 against time (μs)

When HA-V01 is plotted against time, the plot is the following.

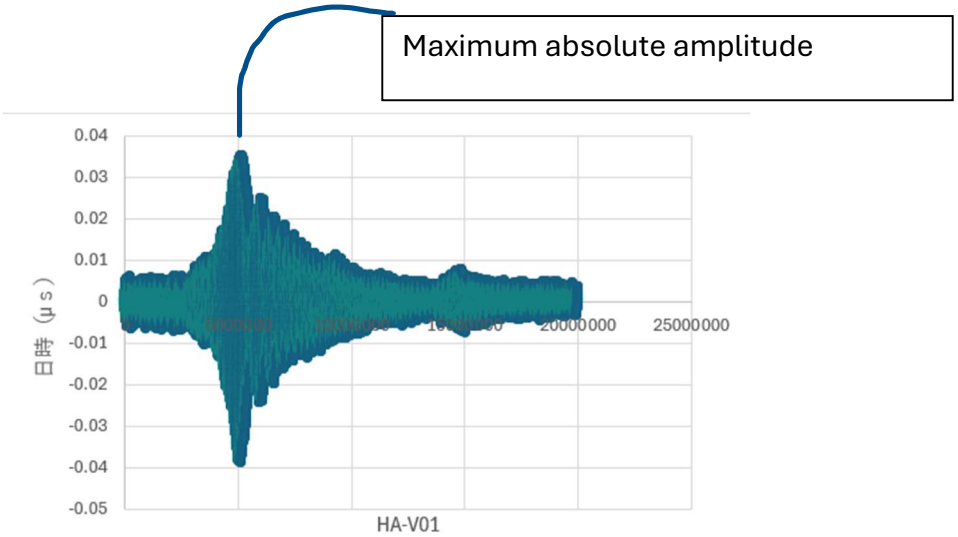


Figure 3 HAV01 against time (μs)

The maximum amplitude is where critical/resonant speed and frequency occurs.

1. To derive the critical frequency from the experimental data(waveforms) from horizontal motion and vertical motion, we used peaks from V01 (horizontal motion data) and V02 (vertical motion data), respectively .

   For V01:

   Peak 1 : V01 0.0035 t1 5151550 μs
   Peak 2: V01 0.0035 t2 5225900 μs

   $$\Delta t = t_1 - t_{2=}74350\mu s$$

   T [s]= Δt measured in seconds =0.07435 s
   Critical frequency=f= 1/T= 13.45 Hz
   Critical speed/angular velocity for horizontal motion $\omega = \frac{2\pi}{T} = 84.5\ rad/s$

   Use similar method for V02:
   Peak 1: V02 0.025 t1 2762800
   Peak 2 V02 0.0242 2830600
   $T[s] = \frac{\Delta t[\mu s]}{1000000\frac{\mu s}{s}} = \frac{t_1 - t_2}{1000000}$ =0.0678s
   $critical\ frequency\ f = 1/T$=14.75 Hz
   Critical speed/ angular velocity for vertical motion $\omega = \frac{2\pi}{T}$=92.7 rad/s

d) To do theoretical calculation of the spring constant, natural frequency ω, and resonant frequency values, we use the following values for the values for the mass M of the rotating disk and the dimensions of the shaft are used from the measurements of the experimental equipment.

- Distance between fulcrums: $l = 693$ mm
- Distance from motor side endpoint: $a = 364$ mm
- Shaft radius: $r = 6$ mm
- Moment of inertia of area: $I = \frac{\pi r^4}{4}$
- Young's modulus of rotating shaft (steel): $E = 206$ GPa
- Poisson's ratio of rotating shaft (steel): $\nu = 0.3$
- Density of rotating shaft (steel): $\rho = 7.86$ g/cm^3
- Disc mass: $M = 4768$ g

The principle of the rotating shaft is below.

The rotational motion of a straight axis supported at both ends: when the rotational speed takes a certain value, rotational motion in which the two centerlines do not coincide. i.e the centerline of the axis swings from the centerline of rotation, occurs.

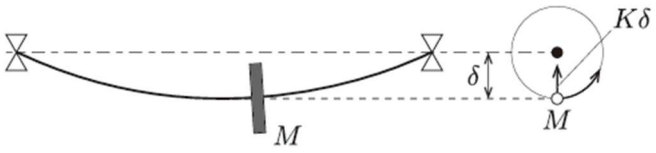


Figure 4 Rotating shaft with a swinging disk

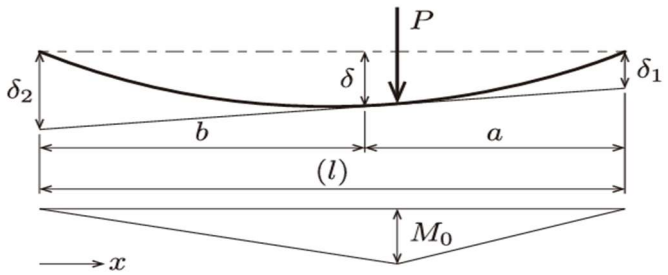


Figure 5 Distribution of deflection and bending moment when a concentrated load is applied to a simply supported beam.

From the textbook (Figures 4 and 5), Critical speed is $\omega = \sqrt{\frac{K}{M}}$ and the spring constant K of the shaft is obtained as follows from beam theory:

$$K = \frac{P}{\delta} = \frac{3EIl}{a^2b^2}.$$

(Equation 1)

Calculate these values:

b= l-a = 329 mm =0.329 m

I= $1.018\times 10^{-9}m^4$

From Equation 1 and the values listed above, K=$3.04 \times 10^4 N/m$

The theoretical critical speed value is $\omega = \sqrt{\frac{K}{M}}$ = 79.84 s.

e) Compare experimental data with theoretical value

1. Why do the critical frequencies differ between the vertical and horizontal directions?

   The critical frequency f_c for horizontal motion = 13.45 Hz, while f_c for vertical motion =14.75 Hz.

   For rotating machinery, the critical speed corresponds to a natural frequency. The difference between the critical frequency for different motions suggests that the rotor-bearing system is not perfectly symmetric. The bearing stiffness or support stiffness is larger in the vertical direction, giving a higher critical frequency there.

   For a rotating shaft, different critical frequencies in the vertical and horizontal directions usually indicate that the shaft-bearing-support system is anisotropic, meaning its stiffness is not the same in all directions.

   The critical frequency is related to the natural frequency of the rotor system:

   $$\omega_n = \sqrt{\frac{k}{m}}$$

   where $k$is the equivalent stiffness and $m$is the effective rotor mass.

   Since the horizontal critical frequency is higher,

   $$\frac{k_h}{k_v} = \left(\frac{14.75}{13.45}\right)^2 \approx 1.20$$

   This means the horizontal stiffness is about 20% greater than the vertical stiffness.

This happens in a rotating shaft due to:

1. Gravity reduces the effective vertical stiffness

The rotor weight acts vertically downward, causing static deflection of the shaft. This preload can make the rotor more compliant in the vertical direction, lowering the vertical natural frequency.

2. Bearing stiffness differs by direction

Real bearings rarely have identical stiffness in all directions. Rolling-element bearings, journal bearings, and their housings often exhibit different horizontal and vertical stiffnesses.

3. Support structure flexibility

The bearing pedestals, frame, or mounting structure may deform more easily in one direction than the other. If the support is softer vertically, the vertical critical speed decreases.

4. Manufacturing and alignment imperfections

Small asymmetries such as:

- bearing misalignment,
- uneven support conditions,
- shaft bow,
- non-uniform foundation stiffness,

can split what would otherwise be a single critical frequency into two nearby frequencies.

The difference between 13.45 Hz and 14.75 Hz is approximately

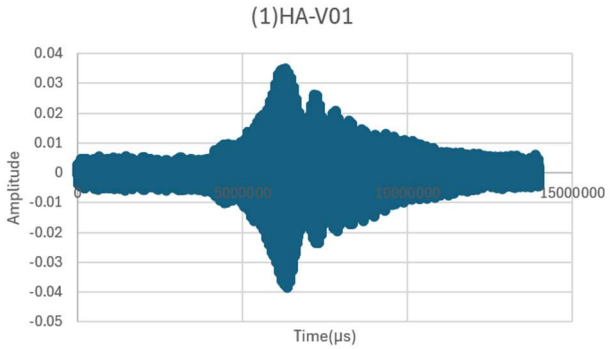
$(14.75-13.45)/14.75\times100\approx8.8\%$

An 8–10% separation is quite common in experimental rotor dynamics and usually indicates unequal support stiffness rather than a modeling error.

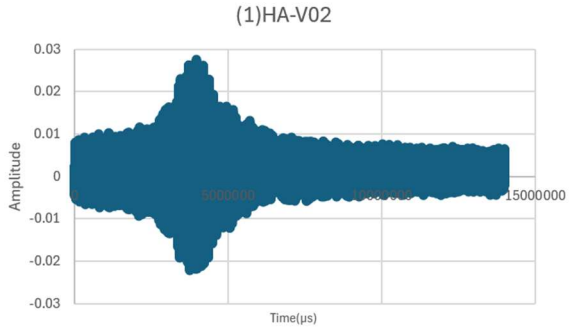
In summary,

The critical frequencies in the vertical and horizontal directions differ because the rotor-bearing-support system exhibits different stiffness characteristics in the two directions. The lower vertical critical frequency (13.45 Hz) suggests a lower effective stiffness due to gravitational loading and support flexibility, while the higher horizontal critical frequency (14.75 Hz) indicates greater lateral stiffness. Consequently, the shaft reaches resonance at different rotational speeds in the two directions.

2. 2nd trial of the Rotary Shaft experiment
   Same method.
   Acquisition of data:
   The graph of V01 amplitude vs time



The graph of V02 amplitude vs time



2. To derive the critical frequency from the experimental data(waveforms) from horizontal motion and vertical motion, we used peaks from V01 (horizontal motion data) and V02 (vertical motion data), respectively.
3. To derive the critical frequency from the experimental data(waveforms) from horizontal motion and vertical motion, we used peaks from V01 (horizontal motion data) and V02 (vertical motion data),

For V01:

Peak 1 : V01 0.0349 t1 6381900 μs

Peak 2: V01 0.0316 t2 6461850 μs

T [s]= Δt measured in seconds = $T[s] = \frac{\Delta t[\mu s]}{1000000\frac{\mu s}{s}} = \frac{t_1 - t_2}{1000000}$ =0.07995 s

Critical frequency=f= 1/T= 12.508 Hz

Critical speed/angular velocity for horizontal motion $\omega = \frac{2\pi}{T} = 78.6\, rad/s$

Use similar method for the vertical motion data V02:

$T[s] = \frac{\Delta t[\mu s]}{1000000\frac{\mu s}{s}} = \frac{t_1 - t_2}{1000000} = 0.067$s

$critical/resonant\ frequency\ f = 1/T$=14.925 Hz

Critical speed/ angular velocity for vertical motion $\omega = \frac{2\pi}{T}$=93.77 rad/s

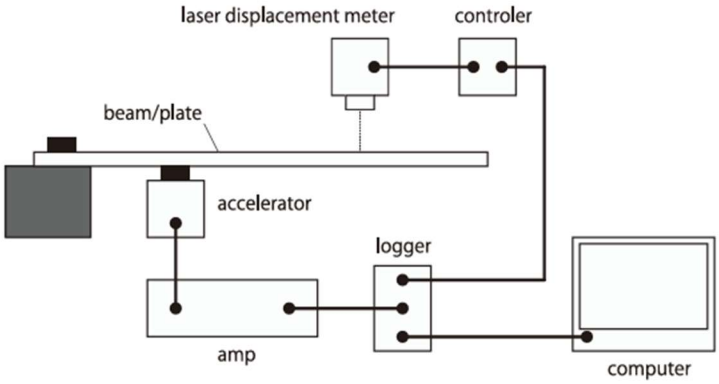
The difference between critical/resonant frequency values between horizontal and vertical motion is due to the same reason as in the discussion regarding the first trial.

2. Vibration of a Flat Plate

Experimental task

1. Find natural frequencies from the lowest to the fourth order
2. Numerical solution of natural frequencies by finite element simulation
3. Compare experimental results with numerical solutions
4. Illustrate vibration modes up to 4th order by numerical solution

Experimental equipment



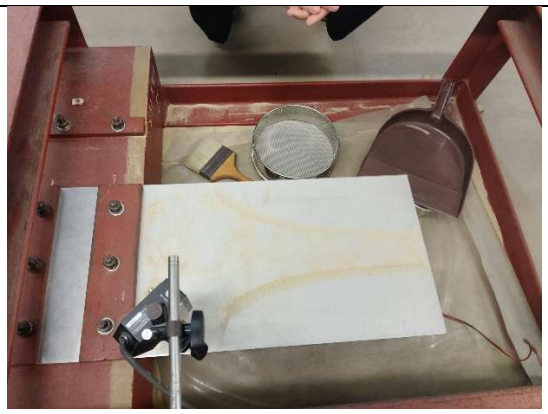
Experimental method

1. Vibration frequency is controlled by the frequency dial on the oscillator to vibrate the test object.
2. Input oscillator signals to a logger and observe them on a computer
3. Sprinkle a thin layer of sand on the test object and determine the resonance frequency from the pattern of the sand.
4. Input the output of the laser displacement meter to the logger and observe it on the computer.

a) Acquired data

The plate specifications: Width b = 300 mm, Length l = 500 mm,
Thickness t = 3.22 mm, Made of aluminum

The resonant frequencies, corresponding resonant vibration modes and corresponding mode shape of the beam obtained in the experiment are in the table below until 4th mode.

Mode	Resonance Frequency [Hz]	Mode shape
1st	9.5	
2nd	33.2	
3rd	56.4	

4th	111.2	

Numerical solution: Finite element method (FEM) simulation

The finite element method is one of the numerical analysis methods for partial differential equations, in which a small region called an element is combined to represent a computational target. Here, we assume a linear elastic body discretised by hexahedral primary elements.

In the FEM method, a continuum body is discretised into mesh. We used the FEM method to solve the solution to Newton's 2nd law. The finite element discretized equation of motion is expressed below

$$M\ddot{x} + kx = 0$$

Here, M is a mass matrix, K is a rigidity matrix, $\ddot{x}$ is a node degree of freedom acceleration vector in finite element discretization, and x is a node degree of freedom displacement vector in finite element discretization.

When the mass point is oscillated while maintaining a constant vibration shape, the displacement and acceleration can be considered by

$$\boldsymbol{x} = \boldsymbol{u}\sin(\omega t + \theta)$$
$$\ddot{\boldsymbol{x}} = -\omega^2 \boldsymbol{u}\sin(\omega t + \theta)$$

Here, t is the time, ω is the angular velocity, θ is the initial phase, and u is the coefficient vector corresponding to the vibration form.

$$(-\omega^2 \boldsymbol{M} + \boldsymbol{K})\,\boldsymbol{u}\sin(\omega t + \theta) = 0$$
$$(-\omega^2 \boldsymbol{M} + \boldsymbol{K})\,\boldsymbol{u} = 0$$

$$M^{-1}Ku = \omega u$$

Leading to solving the general eigenvalue problem.

$$\boldsymbol{Ku} = \lambda \boldsymbol{Mu}$$

Here, $\lambda = \omega^2$ is the eigenvalue and u is the eigenvector.

We will use the FEM simulator on a flat plate to search for eigenvalues which are calculated resonant frequencies, until 4th mode. For other physical properties and dimensions of the flat plate, the following values from prior measurements shall be used.

Flat plate width: b = 300 mm=0.3 m

Flat plate length: l = 500 mm =0.5 m

Plat plate thickness: t = 3.2 mm =0.0032 m

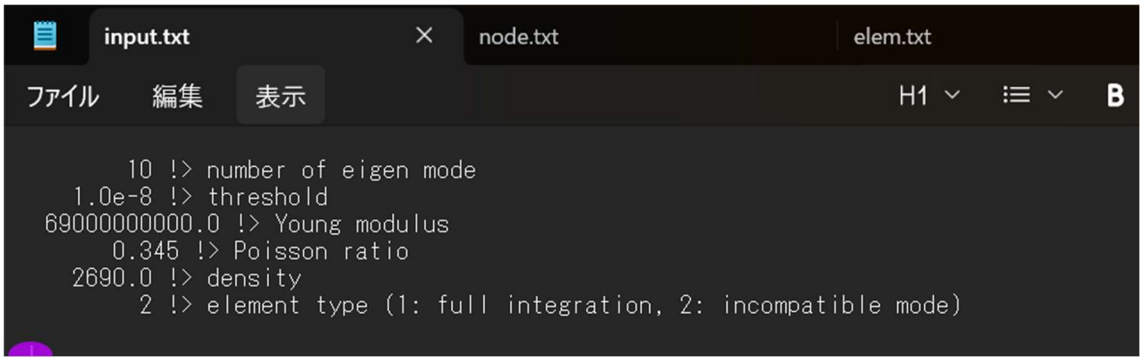
Young's modulus of flat plate (aluminum): E = 69 GPa =69000000000 Pa

Poisson's ratio of flat plate (aluminum): ν = 0.345

Density of flat plate (aluminum): ρ = 2.69 g/cm3

The Finite Element simulator is run on WSL or Ubuntu using a gfortran.f90 mesher (Fortra). The simulator was installed from Github and is stored in the folder/directory monolis_eigen. Then we visualise the mode patterns/shape of the plate with a certain mode number and corresponding eigenvalue/resonant frequency using PARAVIEW software.

After installing and compiling necessary packages, we first substituted the input values of the plate in accordance with experimental values in notepad file stored in monolis_eigen/example/plate folder.



Then, to run the FEM simulator, use the following command

Compiling the Mesher

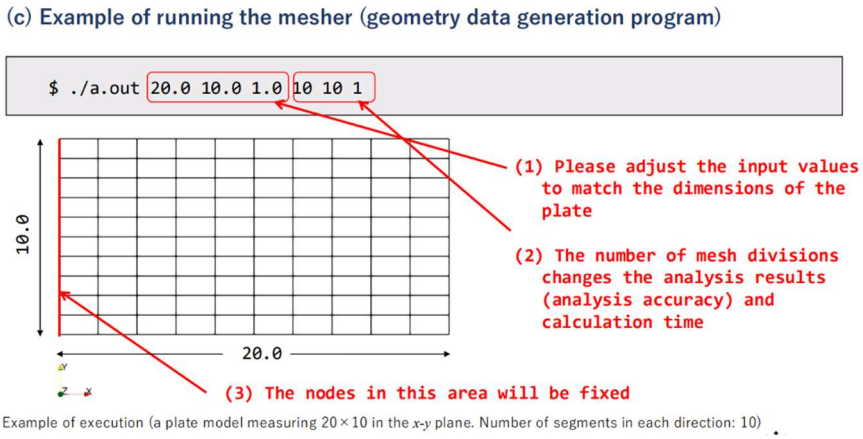
```
~$ cd monolis_eigen/example/plate
~/monolis_eigen/example/plate$ gfortran mesher.f90
```

Next,

Run the mesher according to this example

./a.out {x-direction length} {y-direction length} {z-direction length} {number of xdirection subdivisions} {number of y-direction subdivisions} {number of z-direction subdivisions}

When executed with these parameters, node.txt, elem.txt, and bc.txt are generated.



For adjusting the input values to match the dimensions of the plate,

The first three numbers (fixed values) in the $./a.out command are

$ ./a.out l b t [mesh division or degree of freedom]

This becomes

$./a.out 0.5 0.3 0.0032

since in the experimental data l=0.5 m, b=0.3 m and 0.0032 m. (using consistent units)

Next, we vary the different degrees of freedom. The number of mesh divisions changes the analysis results(analysis accuracy) and calculation time.

a) Using Mesher values 100 60 1

```
~/monolis_eigen/example/plate$ gfortran mesher.f90
~/monolis_eigen/example/plate$ ./a.out 0.5 0.3 0.0032 100 60 1
```

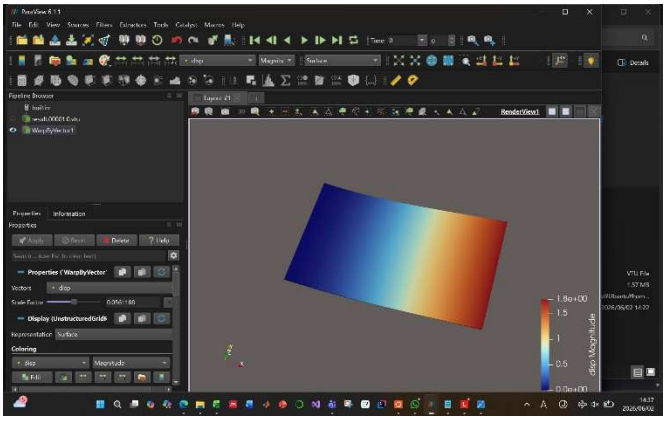
The output of the FEM simulation are the following eigenfrequencies for each of the modes, along with output elapse time and total elapse time.

Mode No	Freq. [Hz]
1	1.08642E+01
2	3.91373E+01
3	6.74102E+01
4	1.30715E+02
5	1.85796E+02
6	2.12849E+02
7	2.61806E+02
8	3.04172E+02
9	3.77715E+02
10	4.61254E+02

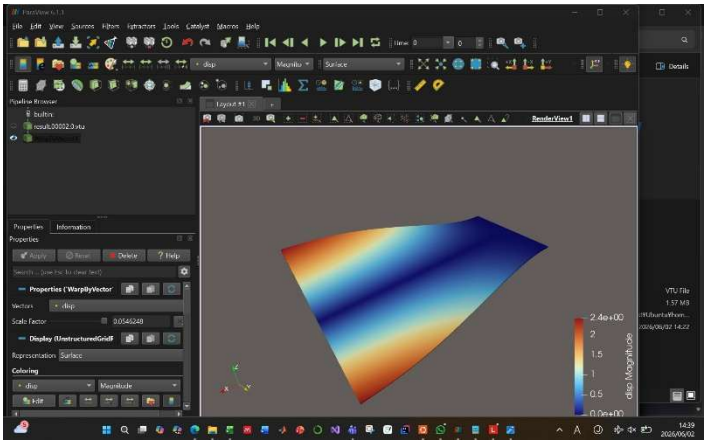
- output elapse time: 5.275E-01 [sec]
- total elapse time: 6.028E+01 [sec]

To verify and compare the calculated mode shapes with experimental results, the numerically simulated plate were visualised in Paraview for each of the modes until 4th mode.

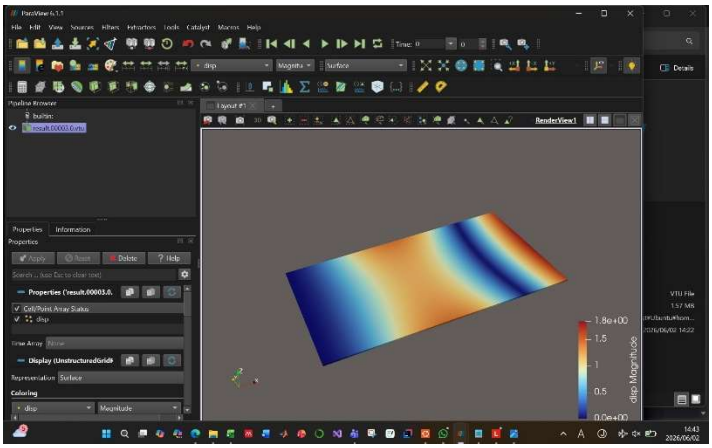
1st mode:



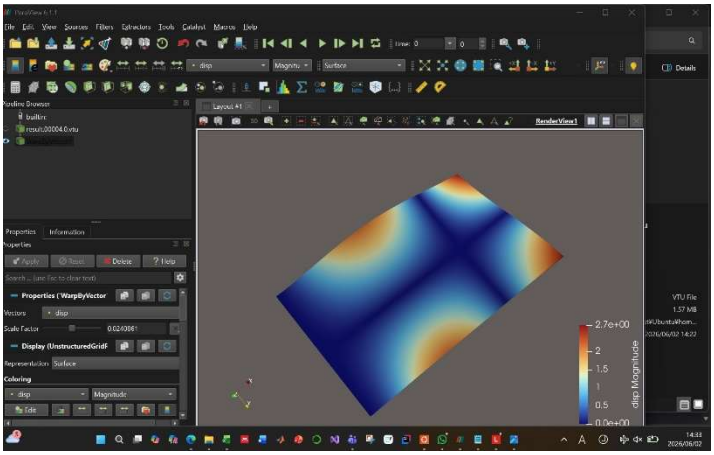
2nd mode:



3rd mode:



4th mode:



The mode patterns show a resemblance to the experimentally obtained mode shapes.

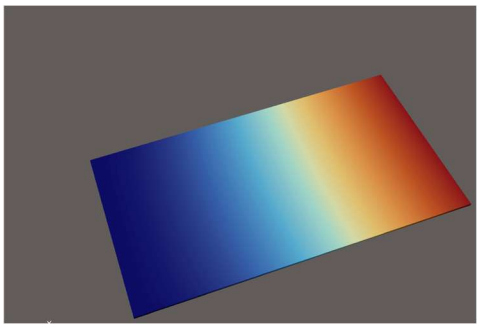
b) Next, I modified the results by increasing the mesh division values to improve accuracy. However, the computational time (total elapse time) increases.
Using Mesher values
150 90 1

```
~/monolis_eigen/example/plate$ gfortran mesher.f90
~/monolis_eigen/example/plate$ ./a.out 0.5 0.3 0.0032 150 90 1
7482
```

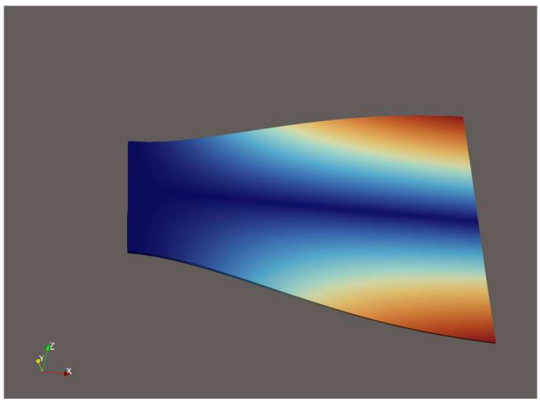
The output mode frequencies are as below. Compared to the first mesher value, the numerical frequency of the 1st mode gradually became closer to the experimentally obtained resonant frequency of the first mode: 9.5 Hz.

```
Mode No     Freq. [Hz]
      1   1.08637E+01
      2   3.91248E+01
      3   6.74073E+01
      4   1.30672E+02
      5   1.85793E+02
      6   2.12886E+02
      7   2.61731E+02
      8   3.04172E+02
      9   3.77628E+02
     10   4.60450E+02
 - output elapse time:  2.327E+00 [sec]
 - total elapse time:  7.933E+01 [sec]
```

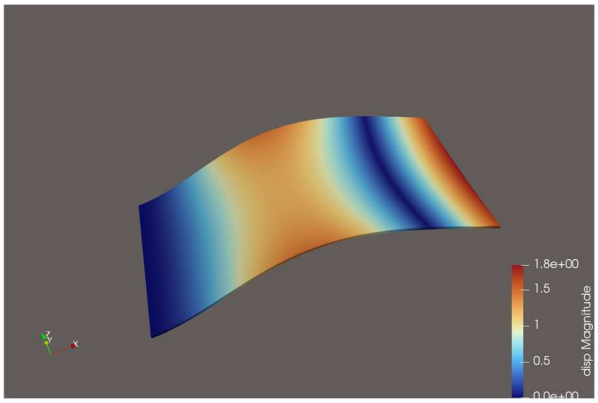
1st mode visualised:



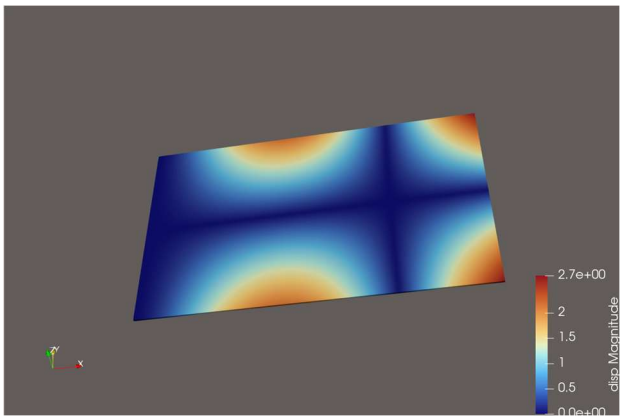
2nd mode:



3rd mode:



4th mode:



Next, decreasing the mesh division number into 90 50 1. As the mesh number decreases, the sensitivity decreases. The 1st mode frequency is farther to the experimentally obtained value. Conversely compared to the second tested mesh division numbers which are higher in value, the total elapsed time for calculation decreases.

```
Mode No    Freq. [Hz]
      1   1.08644E+01
      2   3.91413E+01
      3   6.74119E+01
      4   1.30729E+02
      5   1.85791E+02
      6   2.12810E+02
      7   2.61831E+02
      8   3.04133E+02
      9   3.77755E+02
     10   4.58324E+02
 - output elapse time:  7.309E-01 [sec]
 - total elapse time:  6.307E+01 [sec]
```

The optimal value for the mesh division seems to be 100 … … (shortest elapsed time), so I change the mesh division from 100 60 1 to 100 80 1.

Using the following command:

```
$ ./a.out 0.5 0.3 0.0032 100 80 1
```

The numerical solutions of the frequencies of each mode are:

```
 Mode No    Freq. [Hz]
       1   1.08638E+01
       2   3.91298E+01
       3   6.74079E+01
       4   1.30697E+02
       5   1.85798E+02
       6   2.12887E+02
       7   2.61778E+02
       8   3.04186E+02
       9   3.77693E+02
      10   4.60580E+02
  - output elapse time:  1.286E+00 [sec]
  - total elapse time:  6.791E+01 [sec]
```

-Other

Q1: The numerically solutions differ from experimental data. This may be due to

1. Boundary Conditions

This is usually the largest source of error.

In FEM, the plate is often assumed to be:

- Perfectly clamped
- Simply supported
- Free-free

In the experiment:

- Clamps are not perfectly rigid.
- Bolts may loosen slightly.
- Supports may introduce compliance.

Since natural frequency depends on stiffness,

$$f_n \propto \sqrt{\frac{k}{m}}$$

a less rigid support reduces the effective stiffness and lowers the measured frequency.

2. Material Properties are Not Exact

   The actual plate may have:

   a) Manufacturing variations

b) Residual stresses
c) Material anisotropy
d) Temperature effects

3. Measurement errors
4. Geometry Errors
   The real plate dimensions may differ slightly from the CAD model.
   Examples:

- Thickness variation
- Corner fillets
- Holes
- Surface coatings
  For plates, thickness is especially important because bending stiffness is proportional to

$$D \propto h^3$$

where $h$is plate thickness.
A small thickness error can produce a significant frequency shift.

5. Mesh Quality and Numerical Errors
   An FEM model can give inaccurate results if:

- Mesh is too coarse
- Element type is inappropriate
- Convergence study was not performed
  A coarse mesh may underestimate or overestimate stiffness.

The FEM frequencies are consistently higher than experimental frequencies due to:

1. Boundary conditions are less rigid than assumed.
2. Accelerometers or fixtures add mass.
3. Plate thickness is slightly smaller than modeled.
4. Experimental setup.

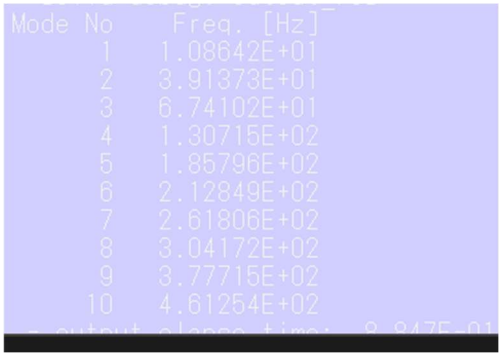
Q2: If necessary, examine the sensitivity of the numerical solutions (the influence of factors such as the number of mesh elements and material constants on the results)

The mesh division numbers and material constants are already discussed previously. Next, the effect of the number of nodes were investigated. Previously, the number of nodes used for all the FEM calculations above was 16 362.

If the number of nodes is lower , e.g 5 000. Using the same material constants and mesh number

$./a.out 100 60 1

It seems that the result frequencies are still the same.



Mode No	Freq. [Hz]
1	1.08642E+01
2	3.91373E+01
3	6.74102E+01
4	1.30715E+02
5	1.85796E+02
6	2.12849E+02
7	2.61806E+02
8	3.04172E+02
9	3.77715E+02
10	4.61254E+02

This may be due to:

- The mesh is already sufficiently refined (converged solution)

- The FEM model uses exact beam shape functions